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Assignment - 3

Course Name :- Database Management System

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Section-A

1. what is SQL? why is it important in DBMS?

→ Structured Query Language is a standard programming language used to store, manage and retrieve data from a database

It is mainly used to communicate with a DBMS like MySQL, oracle etc.

It is very important in DBMS because:

- Easy Data Management
- Fast Data Retrieval
- Standard language
- Data Security & Control
- Efficient Data handling

2. what is difference between DDL and DML?

DDL	DML
Data Definition language	Data Manipulation language
• It defines structure of a database	• It Manages data inside tables.
• It Focuses on Schemas	• It focuses on Records
• It cannot be Rollback	• It can be Rolled Back
• Commands:	• Commands:
→ CREATE	→ INSERT
→ ALTER	→ UPDATE
→ DROP	→ DELETE
→ TRUNCATE	→ SELECT

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3. what is a JOIN? Explain its types.

→ A join in SQL is used to combine rows from two or more tables based on a related column between them (usually a primary key and foreign key relationship).

Types of JOIN in SQL:-

1. INNER JOIN

Returns only matching records from both tables.

2. LEFT JOIN

Returns all records from the left table and matching records from the right table. if no match, NULL is shown.

3. RIGHT JOIN

Return all records from the right table and matching records from the left table.

4. FULL JOIN

Returns all records from both tables, matching where possible

5. CROSS JOIN

Return all possible Combinations of both tables.

6. SELF JOIN

A table is joined with itself.

4. what is difference between ~~WHERE~~ GROUP BY and HAVING clause?

→ The GROUP BY clause is used to group rows that ~~s~~ have the same values in one or more columns. It is mainly used

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GROUP BY to count students

with aggregate functions like:

- COUNT ()
- SUM ()
- AVG ()
- MAX ()
- MIN ()

The HAVING clause is used to filter groups created by GROUP BY.

It is like WHERE, but:

- WHERE filters rows before grouping
- HAVING filters groups after grouping

5. what is a subquery?

→ A subquery is a query written inside another SQL query. The inner query is executed first, and its result is used by the outer query.

6. what is relational algebra?

→ Relational algebra is a formal mathematical language used in DBMS to query and manipulate data in relational databases. It is a set of operations used to retrieve and work with data from tables

Basic operations of Relational Algebra are:-

1. SELECT (σ)
2. PROJECT (π)
3. UNION ($\cup$)
4. SET DIFFERENCE ($-$)
5. JOIN ($\bowtie$)

... using GROUP BY to count students

7. what is difference between WHERE and HAVING.

→ Both WHERE and HAVING are used to filter data in SQL, but they work at different stages.
WHERE clause works on rows and used before GROUP BY and it cannot use aggregate functions like SUM(), AVG() directly.
HAVING clause works on groups and used after GROUP BY and it can use aggregate functions like SUM(), AVG() etc

Section-B

→ write an SQL query to retrieve all students with marks >75.

- SELECT * FROM students
  WHERE marks >75;

→ write a query using JOIN to combine student and course tables.

- Assuming
  student (student_id, name, course_id)
  course (course_id, course_name)
  query:-
  SELECT student.name, course.course_name
  FROM student
  INNER JOIN course
  ON student.course_id = course.course_id;

→ write a query using GROUP BY to count students per department.

- SELECT department, COUNT(*) As total_students
  FROM students
  GROUP BY department;

Section-C

→ A company database stores employee and department data. write a query to find employees in a specific department.

- Assume
  Employee(emp_id, name, department_id)
  Department(department_id, department_name)

  Query:-
  SELECT Employee.emp_id, Employee.name,
  Department.department_name
  FROM Employee
  INNER JOIN Department
  ON Employee.department_id = Department.
  department_id
  WHERE Department.department_name = 'CSE';

→ An e-commerce system stores orders. Retrieve total sales per Customer.

- Assuming we have table:
  orders(order_id, customer_id, amount)

Query

```
SELECT customer_id, SUM(amount)
 As total_sales
FROM orders
GROUP BY customer_id;
```

→ A university needs top 3 students. write a query.

- SELECT student_id, name, marks
  FROM students
  ORDER BY marks DESC
  LIMIT3;

Section-D

→ write an SQL Query for any real life-system and explain logic.

- Example:- Online Shopping System
  In this find total amount spent by each customer, but show only those customers who have spent more than 5000.

Query:-

```
SELECT customer_id,
 SUM(amount) As total_spent
 FROM orders GROUP BY
  customer_id HAVING
   SUM (amount) > 5000;
```

From orders, we take data from the orders table and group all orders belonging to the

same customer and calculate total spending of each customer and filter only those customers whose total spending is greater than 5000.

→ write a JOIN query and mention one use case.

- Ex:- Student & Course Database
  → Retrieve Student names along with the courses they are enrolled in.

Query

```
SELECT s.student_name,
       c.course_name
FROM
    student s
INNER JOIN
    course c
ON
    s.course_id = c.course_id;
```

use case:- In a University system:
→ To display which student is enrolled in which course then this is very useful.

→ write a GROUP BY query and explain its output.
Ex:- E-commerce System:- Find total number of orders placed by each customer

SQL Query:-

```
SELECT customer_id,
   COUNT(order_id) As total_orders
FROM
   orders
GROUP BY
   customer_id;
```

output:

customer_id	total_orders
101	5
102	2
103	7

Here Each row represents one customer and total_orders shows how many orders that customer has placed.